

Edwin Julian

Front-End Engineer (React · Next.js · TypeScript)

Remote (Open to Global, **UTC+7**) · English (Professional)

edjulsin.dev · github.com/edjulsin · edjulsin@gmail.com

Summary

Front-End Engineer with **5+ years of experience** building **production-grade, high-performance web applications**, including **real-time data visualization** and **scalable content platforms**. Strong in **React, Next.js, TypeScript, and D3.js**, with a focus on **architecture, performance, and complex UI interactions**. Comfortable **working autonomously**, owning features **end-to-end**, and collaborating with cross-functional teams in **remote, product-driven environments**.

Selected Projects

FINNS — Booking Engine & Interactive Seat Chart ([Live](#))

Next.js, TypeScript, D3.js, Tailwind CSS, Radix UI, Figma

- Built a high-fidelity booking engine focused on **conversion-driven seat selection UX** using **selective information display (carousel and collapsible components)** to reduce **cognitive overload** and improve **decision-making flow**.
- Developed a **zoomable, map-like seat chart (D3)** with smooth transitions for interactive elements to reinforce **feedback** and improve **perceived responsiveness**, enabling **intuitive navigation** and **mobile-friendly experience**.
- Engineered a **non-blocking, adaptive UI system** with **dark theme and focus mode** to reduce **booking friction** and **cognitive overload** across devices.
- Implemented **smart selection features** (best available, filtering) and **shareable booking state via URL**.

Krypton — Real-Time Trading & Charting Platform ([Live](#))

React, RxJS, D3.js, HTML5 Canvas, Webpack, Express

- Architected and built a **TradingView-class real-time charting system** from first principles, handling high-frequency market data and complex user interactions.
- Designed **RxJS-based streaming pipelines** for WebSocket data ingestion, implementing **reconnection and retry strategies** alongside **lazy loading, multi-chart synchronization, and multi-timeframe analysis**.
- Implemented a **high-performance canvas rendering pipeline** with **programmatic and gesture-driven pan/zoom**, responsive layouts, and mobile optimization.
- Developed a **configurable charting framework** (scales, indicators, themes, overlays) and a custom **color picker & popover system** for advanced visualization control.

Katt — CMS & Visual Blog Editor ([Live](#))

Next.js, TypeScript, Supabase, Tailwind CSS, Radix UI

- Led the architecture and development of a **production-grade CMS**, owning the system across **frontend, data modeling, authentication, authorization, testing, and deployment**.
 - Designed and built a **drag-and-drop visual editor** with a custom layout engine, image crop/resize tools, and responsive grid generation, enabling fully customizable and **SEO-optimized** pages.
 - Optimized **image-heavy rendering paths** using **Next.js Image, ISR, and Core Web Vitals** tuning, significantly improving load performance without sacrificing visual quality.
 - Designed **secure, multi-role collaboration workflows** using **Supabase Row Level Security (RLS)** and fine-grained permission models.
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Professional Experience

Product Engineer | FINNS Beach Club (Self-Initiated Product)

January 2026 – Present

- Design and build a **booking engine product** centered around an **interactive seat chart** for high-conversion user flows.
- Developed a **zoomable, map-like interface** (D3) enabling intuitive navigation and **mobile-friendly seat selection**.
- Engineered a **non-blocking, adaptive UI system** to maintain uninterrupted interaction across all screen sizes.
- Built **smart selection features** (best available, filtering, shareable state) to reduce friction and improve decision speed.
- Designed a **focused visual system** (dark UI, dynamic labeling) to reduce **cognitive load**.
- Implemented **multi-date booking, per-spot add-ons, and real-time availability** in a **unified flow**.
- Built a **structured, toggle-based checkout** for clear, non-overwhelming order review.

Front-end / Full-Stack Engineer | Katt Interior (Remote)

January 2025 – January 2026

- Architected and delivered a **custom CMS platform** from zero to production, collaborating closely with designers and business stakeholders.
- Designed and implemented an easy-to-use **drag-and-drop content editor** with persistent layouts, crop/resize tooling, and SEO-friendly output.
- Improved **page load speed and Core Web Vitals** by **downscaling large source images (e.g., 5000px → max 1920px)** and applying **responsive, optimal sizing with Next.js Image**, reducing image payloads by **50%+ on average** without visible quality loss and significantly improving **LCP** on media-heavy pages.
- Implemented **role-based access control and secure data isolation** using **Supabase RLS**.
- Established **robust test coverage** with **Vitest (unit)** and **Playwright (browser/e2e)** to ensure critical interactions—**drag-and-drop, resize, and crop**—remain stable and regression-free.
- Applied **on-page and technical SEO best practices** (semantic HTML, metadata, structured data, sitemap, performance optimization) to increase **search visibility and indexability**.

Front-End Engineer | Krypton (Self-Initiated Product)

January 2020 – January 2025

- Designed and built a **real-time financial visualization platform** with complex interaction models and high-throughput rendering.
 - Solved performance-critical problems including **canvas batching, draw-call optimization, viewport scaling, event throttling**, and **lazy-loaded rendering pipelines** to minimize initial load and improve frame stability.
 - Built a **primitive UI component library** (context menus, dropdowns, popovers) to support **chart customization workflows**, with configurable options for **time scales, indicators, themes, and interaction modes**.
 - Implemented **reactive data flows** using **RxJS** to manage streaming updates, backpressure, and UI synchronization.
 - Optimized large crypto data tables by introducing **virtualized lists and incremental filtering/search**, enabling smooth interaction with **hundreds of real-time rows** while reducing memory usage and re-render cost.
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Technical Skills

Frontend & UI Systems

React, Next.js (App Router), JavaScript(ES6+), TypeScript, HTML5 Canvas, RxJS, Jotai, D3.js, Tailwind CSS, SASS, Radix UI, Figma

Backend & Platform

Supabase (PostgreSQL, Auth, RLS, Storage), REST APIs, GraphQL, Express.js, Docker, Vercel

Architecture & Quality

State Management, Real-Time Systems, WebSocket Streaming, Performance Optimization, CI/CD, Git, GitHub Actions, Webpack, Vite, Vitest, Playwright, SEO, Core Web Vitals, WCAG, Biome, ESLint

Education

Full Stack Web Development Certification

freeCodeCamp

January 2019 – January 2020

Bachelor of Economics

Parahyangan Catholic University

August 2010 – March 2016